

AP-E11 Compatible Tetrahedral Cochain Action: Exact Leibniz Algebra, Positive Whitney Hodge Star, Shared Degree Cochain, and a Mesh-Independent Complete-Minor Cell Formula

Route-F derivation ledger

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Abstract

AP-E10 proved that the one-corner forward Skyrme stencil is not a compatible discrete differential product. AP-E11 replaces it by one ordered tetrahedral cochain complex. The Alexander–Whitney front/back cup satisfies the exact Leibniz identity; its target-antisymmetric products are precisely the de Rham cochains of the affine two- and three-minors. Consistent Whitney mass matrices give positive, rigid-rotation-covariant Hodge stars. A radial dressing of the same third-cup family is the exact degree density of the normalized piecewise-affine map. Adding the positive three-cochain Hodge norm closes the finite-grid collapse mode and gives the polyconvex density

$$Q_{R,K}(A) = \frac{1}{2}|A|^2 + \frac{R}{2}|\wedge^2 A|^2 + \frac{K}{2}|\wedge^3 A|^2.$$

Every periodic P1 corrector preserves the means of all three minor orders; Jensen’s inequality therefore proves that the corrector is zero and the cell formula is identical on the six- and checkerboard five-tetrahedron meshes. The production card passes 11/11 exact/mechanical checks, all thirteen unanchored $B = 1$ solves, and the declared translation/triangulation quotient continuum gate. What is closed unconditionally is Gamma convergence to the fixed-degree lower-semicontinuous relaxation. Identification of that relaxation with the unrelaxed classical graph-norm functional remains a separate density theorem; consequently the classical Hessian and determinant variation are not yet authorized.

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1 Why the action must be replaced

Let K_h be an ordered, oriented, shape-regular tetrahedral complex in a bounded Lipschitz domain $\Omega \subset \mathbb{R}^3$. A nodal field is a zero-cochain $n_h = (n_h^0, \dots, n_h^3) \in C^0(K_h; \mathbb{R}^4)$ with $|n_h(v)| = 1$ at every

vertex. AP-E10 sampled all spatial derivatives from one cube corner and multiplied them without a cochain shift. The exact oscillatory counterexample then retained a nonzero discrete two-minor while converging uniformly to a constant. No coefficient calibration can repair that algebraic failure.

The replacement has three requirements. First, $d^2 = 0$ and the product used to form a minor must satisfy a discrete Leibniz rule. Second, the positive quadratic form called a Hodge star must reconstruct affine exterior forms independently of a cube body diagonal. Third, the topological three-cochain must refer to the same normalized-affine map as the finite-grid degree. Whitney forms and finite element exterior calculus provide the appropriate commuting-complex architecture [1, 2, 3]. Recent discrete exterior-calculus work likewise emphasizes that a proposed wedge must be checked against Leibniz rather than inferred from visual locality [4, 5]; those surface results are motivation, not a proof of the present three-dimensional construction.

2 Exact shifted cup algebra

For an increasing simplex $[v_0 \dots v_{p+q}]$, define the front/back Alexander–Whitney product

$$(\alpha \smile_h \beta)[v_0 \dots v_{p+q}] = \alpha[v_0 \dots v_p], \beta[v_p \dots v_{p+q}], \quad \alpha \in C^p, \quad \beta \in C^q. \quad (1)$$

Theorem 2.1 (Differential graded identities). *The simplicial coboundary and (1) obey*

$$d^2 = 0, \quad (2)$$

$$d(\alpha \smile_h \beta) = d\alpha \smile_h \beta + (-1)^p \alpha \smile_h d\beta, \quad (3)$$

$$(\alpha \smile_h \beta) \smile_h \gamma = \alpha \smile_h (\beta \smile_h \gamma). \quad (4)$$

Proof. Expand d as the alternating sum of deleted vertices. Deleting one of the front vertices produces the first term in (3); deleting a back vertex produces the second, including the sign from passing p front degrees. The two copies obtained by deleting the shared vertex cancel. A second deletion cancels in pairs, proving (2). Both sides of (4) evaluate to the product on the three consecutive front, middle, and back blocks. $\square$

Put $a^A = dn_h^A$. The correctly normalized antisymmetric two-cochain is

$$X_h^{AB} = \frac{1}{2!} (a^A \smile_h a^B - a^B \smile_h a^A). \quad (5)$$

On an oriented face [012],

$$X_h^{AB}[012] = \frac{1}{2} \det \begin{pmatrix} n_1^A - n_0^A & n_2^A - n_0^A \\ n_1^B - n_0^B & n_2^B - n_0^B \end{pmatrix} = \int_{[012]} d(I_h n^A) \wedge d(I_h n^B). \quad (6)$$

The factor $1/2!$ is essential: the reference triangle has area $1/2$. Similarly define the fully antisymmetrized family

$$Z_h^{ABC} = \frac{1}{3!} \sum_{\pi \in S_3} \text{sgn}(\pi) a^{\pi(A)} \smile_h a^{\pi(B)} \smile_h a^{\pi(C)}. \quad (7)$$

It is the de Rham three-cochain of the affine third minor.

There is also a distinguished target contraction. If $T = [0123]$, direct expansion gives

$$C_h[n](T) = \epsilon_{ABCD} (n^A \smile_h a^B \smile_h a^C \smile_h a^D)(T) = \det[n_0 \ n_1 \ n_2 \ n_3]. \quad (8)$$

Thus the energy cochains (7) and the topological contraction (8) belong to one differential graded algebra; they are not separate stencils.

3 Positive rotation-calibrated Hodge stars

Let λ_i be the barycentric coordinates of a tetrahedron. For an oriented k -face $\sigma = [i_0 \dots i_k]$, the Whitney form is

$$\mathcal{W}_h(\sigma) = k! \sum_{r=0}^k (-1)^r \lambda_{i_r} d\lambda_{i_0} \wedge \dots \wedge \widehat{d\lambda_{i_r}} \wedge \dots \wedge d\lambda_{i_k}. \quad (9)$$

Define the local consistent Hodge matrix by

$$(M_k^T)_{\sigma\tau} = \int_T \mathcal{W}_h(\sigma) \wedge \star \mathcal{W}_h(\tau), \quad k = 1, 2, 3. \quad (10)$$

Proposition 3.1 (Positivity and rotation calibration). *For every nondegenerate tetrahedron, M_k^T is positive definite. Under a rigid rotation $Q \in SO(3)$, the naturally identified matrix satisfies $M_k^{QT} = M_k^T$. If f, g are affine, then*

$$(df)^T M_1^T (df) = \int_T |df|^2, \quad (11)$$

$$(X_h^{fg})^T M_2^T (X_h^{fg}) = \int_T |df \wedge dg|^2, \quad (12)$$

$$(Z_h^{fgh})^T M_3^T (Z_h^{fgh}) = \int_T |df \wedge dg \wedge dh|^2. \quad (13)$$

Proof. Equation (10) is a Gram matrix. Linear independence of the local Whitney basis makes its nullspace trivial. A rigid rotation preserves volume and the Euclidean inner product on exterior forms, giving covariance. The commuting identity $\mathcal{W}_h d = d\mathcal{W}_h$ reconstructs every constant differential of an affine function. Equations (6) and (7) then give (11)–(13). $\square$

This consistent star is deliberately not replaced by a diagonal circumcentric formula; diagonal positivity can fail on non-Delaunay tetrahedra. Rotational calibration here means exact affine reconstruction, not an empirical average of body diagonals.

4 The shared cochain is the normalized-affine degree

On a tetrahedron write

$$v_T(\lambda) = \sum_{i=0}^3 \lambda_i n_i, \quad u_T = \frac{v_T}{|v_T|}, \quad \lambda \in \Delta^3. \quad (14)$$

Fix $-1/3 < \epsilon < 1$ and impose $n_i \cdot n_j > \epsilon$ for all vertex pairs of every tetrahedron. Then

$$|v_T|^2 \geq \epsilon + (1 - \epsilon) \sum_i \lambda_i^2 \geq \frac{1 + 3\epsilon}{4} > 0. \quad (15)$$

The production action uses $\epsilon = -0.32$, for which the last lower bound is 10^{-2} ; every relaxed solution is in fact much farther from it.

Let $s_T = \text{sgn det}[x_1 - x_0, x_2 - x_0, x_3 - x_0]$ and define

$$\Omega_h[n](T) = \frac{s_T}{2\pi^2} C_h[n](T) \int_{\Delta^3} \frac{d\lambda}{|v_T(\lambda)|^4}. \quad (16)$$

Theorem 4.1 (Exact degree identity). *If the boundary is the vacuum and (15) holds, the normalized affine pieces glue to a continuous map $u_h : \Omega/\partial\Omega \rightarrow S^3$, and*

$$\sum_{T \in K_h^{(3)}} \Omega_h[n](T) = \deg u_h \in \mathbb{Z}. \quad (17)$$

Proof. On the reference tetrahedron,

$$u_T^* \text{vol}_{S^3} = \frac{\det[v_T, \partial_1 v_T, \partial_2 v_T, \partial_3 v_T]}{|v_T|^4} d\lambda.$$

The determinant is independent of λ and equals $\det[n_0, n_1, n_2, n_3] = C_h[n](T)$. Multiplication by s_T converts the reference orientation to the physical orientation. Summing and dividing by $\text{vol}(S^3) = 2\pi^2$ is the pullback definition of degree. $\square$

This identity is stronger than agreement with one regular value. The code nevertheless checks three regular values and an independent Duffy–Gauss evaluation of (16).

5 Compatible complete-minor action and cell formula

Let $v_h = I_h n_h$ be the P1 affine interpolant. The AP-E11 action is

$$E_h[n_h] = \sum_{T \in K_h} \int_T \left\{ \frac{1}{2} |Dv_h|^2 + \frac{R}{2} |\wedge^2 Dv_h|^2 + \frac{K}{2} |\wedge^3 Dv_h|^2 + m^2(1 - v_h^0) \right\} dx, \quad (18)$$

on the admissible nodal S^3 sector, and $+\infty$ outside it. The production card uses $(R, K, m^2) = (1, 0.35, 0.12)$. Equations (11)–(13) express every derivative term as a positive cochain Hodge norm. The K term is not an unrelated stabilizer: it is the norm of the complete family (7) whose contraction and radial dressing give (16). For a smooth S^3 map its norm is the squared topological-current density.

For a periodic P1 corrector $\varphi : Y \rightarrow \mathbb{R}^4$, define

$$Q_{R,K}^{\text{cell}}(A) = \inf_{\varphi \text{ periodic}} \int_Y Q_{R,K}(A + D\varphi) dx, \quad Q_{R,K}(A) = \frac{1}{2}|A|^2 + \frac{R}{2}|M_2(A)|^2 + \frac{K}{2}|M_3(A)|^2. \quad (19)$$

Theorem 5.1 (Finite- R, K mesh-independent cell formula). *On every conforming periodic tetrahedral mesh,*

$$\boxed{Q_{R,K}^{\text{cell}}(A) = Q_{R,K}(A), \quad \varphi = 0 \text{ is a global minimizer.}} \quad (20)$$

This holds for the uniform six-tet mesh and for both checkerboard five-tet phases, with no mesh-dependent rescaling.

Proof. Periodicity and the null-Lagrangian identities give, for $r = 1, 2, 3$,

$$\frac{1}{|Y|} \int_Y M_r(A + D\varphi) dx = M_r(A). \quad (21)$$

For $r = 1$ this is the mean-zero gradient identity. For $r = 2, 3$, expand each minor and integrate one derivative by parts; periodic boundary terms vanish and mixed derivatives cancel. The function $(F, G, H) \mapsto (|F|^2 + R|G|^2 + K|H|^2)/2$ is convex. Jensen applied to the complete minor vector and (21) proves the lower bound $Q_{R,K}(A)$, attained at zero corrector. This is the standard polyconvex mechanism [6], evaluated here on the actual finite periodic complex rather than invoked as an asymptotic slogan. $\square$

6 Compensated compactness and the precise Gamma theorem

Let $a_h \rightarrow 0$ on a shape-regular mesh family and suppose $\sup_h E_h[n_h] < \infty$. The Dirichlet term bounds v_h in H^1 . Rellich gives $v_h \rightarrow v$ strongly in L^2 after a subsequence. Nodal unit length and the local interpolation estimate

$$\|1 - |v_h|\|_{L^2} \leq Ca_h \|Dv_h\|_{L^2} \quad (22)$$

imply $|v| = 1$ almost everywhere.

The action bounds $M_2(Dv_h)$ and $M_3(Dv_h)$ in L^2 . Because the cochains are actual affine minors, not one-corner products, the distributional null-Lagrangian identities identify their weak limits with $M_2(Dv)$ and $M_3(Dv)$. Equivalently, each third minor satisfies the Piola form

$$\det D(v_h^A, v_h^B, v_h^C) = \frac{1}{3} \operatorname{div} \left((v_h^A, v_h^B, v_h^C) \operatorname{cof} D(v_h^A, v_h^B, v_h^C) \right). \quad (23)$$

This is precisely the differential constraint missing in AP-E10; general div–curl theory explains why such compatibility is indispensable [7].

The normalized current has the exact affine formula

$$J_h = \frac{\det[v_h, D_1 v_h, D_2 v_h, D_3 v_h]}{|v_h|^4}. \quad (24)$$

The uniform gap (15), strong finite- L^p convergence of v_h , and weak L^2 compactness of the third minors identify the weak limit of J_h . In particular its integral and hence degree are closed.

Theorem 6.1 (Unconditional relaxed fixed-degree Gamma theorem). *Fix a degree B , the boundary vacuum, $-1/3 < \epsilon < 1$, and a shape-regular mesh family. In strong L^2 , the discrete functionals (18) Gamma-converge on the degree- B sector to the fixed-degree lower- semicontinuous relaxation of the smooth complete-minor action.*

Proof. Equicoercivity and sector closure were established above. Convex weak lower semicontinuity in $(Dv, M_2(Dv), M_3(Dv))$ gives the liminf. For every smooth S^3 map, nodal interpolation is eventually admissible, preserves degree, and converges in all three Hodge energies by affine exactness. Apply that recovery construction to a smooth sequence defining the fixed-degree lower-semicontinuous relaxation and take a diagonal subsequence. $\square$

The last phrase is important. This theorem does not assert that the relaxed functional equals the naive classical graph-norm functional on every finite- energy Sobolev map. Smooth approximation of manifold-valued Sobolev maps is topology-sensitive [8]; the exact density problem for the simultaneous Dv , $M_2(Dv)$, and $M_3(Dv)$ graph norm is not supplied by the standard theorem. Recent rigorous lattice-skyrmion work gives a useful compactness/liminf/recovery architecture in two dimensions [9], but does not identify this three-dimensional graph-norm relaxation.

7 Numerical verification and quotient gate

The independent card is

`route_f/code/scan_ap_e11_compatible_cochain_action.py`.

It verifies exact integer cochain identities, local Hodge matrices, radial degree integration, periodic cell optimizations, and thirteen unanchored stationary fields. Selected residuals are

check	observed maximum/minimum
d^2 , Leibniz, associativity residual	0
third-cup determinant residual	8.33×10^{-16}
minimum local Hodge eigenvalue	2.485×10^{-2}
rigid-rotation Hodge residual	3.55×10^{-15}
affine one-/two-/three-form reconstruction residual	2.40×10^{-16}
radial cochain versus regular-value degree	3.07×10^{-7}
periodic complete-minor mean residual	1.78×10^{-15}
optimized cell-formula relative residual	2.56×10^{-14}
maximum projected gradient density, 13 cases	1.54×10^{-7}
minimum pair dot product, 13 cases	0.2724

Every case has three-target $B = (1, 1, 1)$. The common sequence extends through $N = 37$, $L = 8$, and $a = 0.222222$. Translation and mesh comparisons use $N = 29$, $L = 7$, and $a = 0.25$. The quotient uses the positive density of the same action, lumped from each exact tetrahedral contribution to its four vertices; its radial comparison is the maximum CDF distance, not a bin-phase-sensitive histogram norm.

gate	observed	threshold	pass
joint-tail energy spread	0.437%	3%	yes
joint-tail centered-radius spread	3.489%	5%	yes
joint-tail radial CDF distance	4.067%	5%	yes
translated-start energy spread	8.33×10^{-12}	1%	yes
translated-start radius spread	7.16×10^{-7}	2%	yes
translated-start CDF distance	5.91×10^{-7}	3%	yes
five-/six-tet energy spread	0.463%	3%	yes
five-/six-tet radius spread	0.752%	5%	yes
five-/six-tet CDF distance	3.556%	5%	yes

Thus

$$\boxed{C_{\text{cell}} = \text{true}, \quad C_{\text{quotient}} = \text{true}, \quad C_{\text{relaxed regulator}} = \text{true}.} \quad (25)$$

The classical-action identification remains

$$\boxed{C_{\text{graph density}} = \text{false}, \quad C_{\text{classical regulator}} = \text{false}, \quad C_{\text{Hessian}} = C_{\text{det}} = \text{false}.} \quad (26)$$

Equation (26) is not a return of the AP-E10 stencil blocker. That blocker is removed. It is a sharper functional-analytic question: either prove fixed-degree smooth density in the complete-minor graph norm, or prove that the selected relaxed minimizer is a regular isolated classical solution. Only then may the same-action Riemann Hessian be assembled; a regulated determinant additionally requires a declared fermion kernel and counterterm prescription.

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